

Data Science Assignment — Web3 Trading Team

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Project: Relationship between Trader Behavior and Bitcoin Market Sentiment

Executive Summary

This report explores the relationship between trader behavior (profitability, volume, win-rate, exposure) and the Bitcoin Fear & Greed Index. Analyses include data cleaning, daily and per-account aggregation, correlation analysis, statistical testing, and trader clustering. Key insights and recommended signals for trading strategies are provided.

Data Sources & Preparation

Datasets: 1. Historical Trader Data (Hyperliquid): Columns include Account, Coin, Execution Price, Size Tokens, Size USD, Side, Timestamp IST, Start Position, Direction, Closed PnL, Transaction Hash, Order ID, Crossed, Fee, Trade ID, Timestamp. 2. Bitcoin Fear & Greed Index: Columns include timestamp, value, classification, date. Processing steps: - Mounted Google Drive and loaded CSVs. - Converted timestamps to datetime and created a 'date' column for daily aggregation. - Ensured numeric types for PnL and sizes, computed Notional where required, and added a boolean 'is_win' indicator. - Aggregated per-day across traders and per-account-per-day aggregates for segmentation and detailed analysis.

Key Aggregated Metrics (sample rows)

date	Total_trades	Total_volume	avg_closed_pnl	Total_pnl	Win_rate	classification
2023-11-14	1045	11,555,430	0.1488	155.50	0.2746	Greed
2024-03-09	6962	39,406,770	25.4188	176,965.5	0.4901	Extreme Greed
2024-10-27	35241	103,946,100	90.5043	3,189,461	0.4516	Greed

Correlation Analysis

Correlation was computed between the numeric sentiment label (mapped: Greed=1, Fear=-1) and daily trading metrics. Key correlations observed (from notebook): - correlation(sentiment_label, total_volume): -0.9904 - correlation(sentiment_label, total_pnl): -0.8823 - correlation(sentiment_label, win_rate): -0.4222 Interpretation: Strong negative correlations indicate that on days labeled as 'Greed' traders' total volume and total PnL tended to be lower in our sample, and win_rate also tended to be lower.

Statistical Testing

Due to class imbalance in the sentiment labels (many 'greed' days and few 'fear' days), classical statistical tests had limited power. After grouping 'neutral' with 'fear' (to increase sample size), the groups still had limited samples: - Greed samples: 4 - Fear+Neutral samples: 2 Given the small sample sizes, formal t-tests and Mann-Whitney tests were not run to completion for reliable inference. Instead, descriptive statistics (means and distributions) and visualizations were used to inform directional insights.

Trader Segmentation (K-Means)

K-Means clustering (k=4) was performed on per-account aggregates (average trades, average PnL, average notional, win_rate). Clusters reveal distinct trader archetypes: - Low-activity, low-risk traders - High-activity, moderate-return traders - High-volume, high-variance traders (higher risk) - Consistent performers with steady win rates A CSV with per-account cluster labels was saved for further drill-down.

Visualizations (saved to Drive)

Key figures generated and saved to your Drive under: -

/content/drive/MyDrive/ds_anubhav_sharma/outputs/total_pnl_by_sentiment.png -

/content/drive/MyDrive/ds_anubhav_sharma/outputs/sentiment_correlation.png

Also saved aggregated CSVs to: -

/content/drive/MyDrive/ds_anubhav_sharma/aggregated_outputs/daily_aggregates.csv -

/content/drive/MyDrive/ds_anubhav_sharma/aggregated_outputs/account_daily.csv -

/content/drive/MyDrive/ds_anubhav_sharma/outputs/account_clusters.csv

Actionable Insights & Recommendations

1. Risk flagging: When sentiment is 'Greed' and aggregated average notional or leverage is rising, raise a systemic risk flag — this period shows lower PnL and higher variability.
2. Contrarian signal: Traders tended to outperform (higher mean PnL) in fearful market regimes; consider exploring contrarian strategies or allocating to historically consistent performers during Fear.
3. Trader scoring: Use cluster labels to build a 'reliability score' — prefer capital allocation to consistent performers and limit exposure to high-variance clusters.
4. Data improvement: Ensure longer time series coverage for Fear-class days to enable robust statistical tests (collect more data or expand date range).

Limitations & Next Steps

- The sample had class imbalance; inferences are directional and require more balanced data. - Some sentiment categories (e.g., 'Extreme Greed') had NaN sentiment_label mapping; ensure consistent mapping or use numeric index where possible (0-100). - Next steps: time-series causality tests (Granger), predictive modeling for next-day PnL sign, and backtesting candidate signals.

Appendix

All code, processed CSVs, and figures are saved within the submission folder structure: ds_anubhav_sharma/ Primetrade_assignment.ipynb, aggregated_outputs/ csv_files/ nnn outputs/ Primetrade_assignment_report.pdf